MIL-C-5/17F
11 September 1972
SUPERSEDING
MIL-C-5/17E
20 September 1966

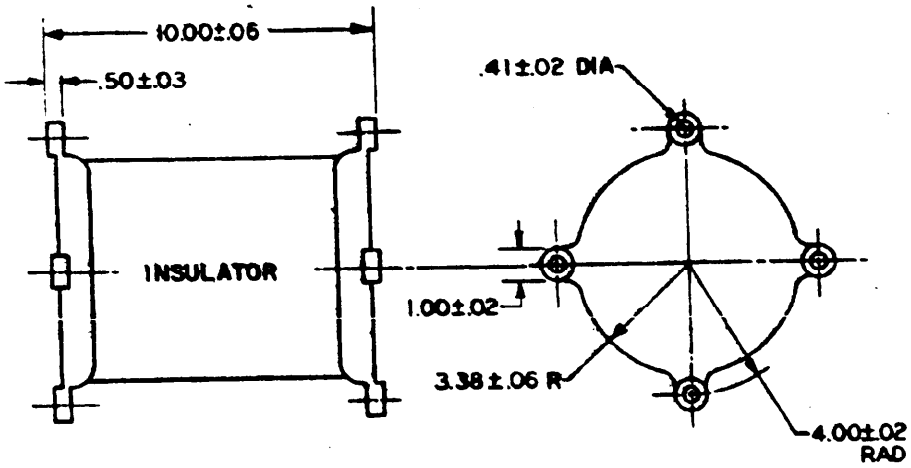
MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, MICA DIELECTRIC,

STYLE CM95

The complete requirements for procuring the capacitors described herein shall consist of this document and the latest issue of Specification MIL-C-5.

STYLE CM95 IS INACTIVE FOR NEW DESIGN AFTER THE DATE OF THIS SPECIFICATION



INCHES	MM	INCHES	MM
.02	.51	1.00	25.40
.03	.76	3.38	85.85
.06	1.52	4.00	101.60
.41	10.41	10.00	254.00
.50	12.70		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Approximate weight 24 pounds (10.88 kg).

FSC 5910

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REQUIREMENTS:

Dimensions: See figure.

Capacitance value: See table I.

Capacitance tolerance: See table I.

Peak working voltage: See table I.

Operating temperature range:

Symbol	Temperature range
M	-55 to +70°C

Vibration grade: Grade 1 (10-55 cps).

Characteristic: See table I.

Symbol	Temperature coefficient P/M/°C	Capacitance drift
B	Not rated	Not rated
D	-100 to +100	±(0.3 percent + 0.1 pf)
F	0 to +70	±(0.05 percent + 0.1 pf)

Dielectric withstanding voltage:
Method 301 of MIL-STD-202:
AC sine-wave potential rms equal to peak working voltage at 100 cycles or less for not less than 5 seconds.

Barometric pressure:
Capacitor, style CM05, covered by this specification sheet is not subjected to a barometric pressure test.

Insulation resistance:
Method 302 of MIL-STD-202, condition A.
At 25°C - 7,500 megohms minimum.
At 70°C - 0 to 5,000 pf -- 2,000 megohms minimum.
Greater than 5,000 pf -- 10 megohm-microfarad.

Capacitance:
Method 305 of MIL-STD-202.

Current rating:
Passage of rated current shall not cause the temperature of the capacitor body's external surface to rise above the ambient temperature by more than 15°C.

Vibration:
Method 201 of MIL-STD-202.

Temperature cycling:
Method 102 of MIL-STD-202, condition D, except the high temperature shall be +70°C.

Dissipation factor, maximum - See MIL-C-5.

Immersion cycling:
Method 104 of MIL-STD-202, condition B.
Dielectric withstanding voltage - Meet initial requirement.
Insulation resistance - 3,000 megohms minimum.
Dissipation factor - Not greater than 150 percent of initial value.
Capacitance change - Shall not exceed ±4 percent of nominal value or 1 pf, whichever is greater.

Shock (medium impact):
Method 205 of MIL-STD-202, condition C (50G).

Moisture resistance:
Method 106 of MIL-STD-202:
Insulation resistance - At least 30 percent of initial requirement.
Dissipation factor - Not greater than 150 percent of initial value.
Capacitance change - Shall not exceed ±5 percent of nominal value or 1 pf, whichever is greater, for characteristic B; ±2 percent of nominal value or 1 pf, whichever is greater, for characteristics D and F.

Temperature coefficient and capacitance drift:
Within the limits specified for the applicable characteristic.

Low temperature storage and life:
48 hours at -55 $^{+0}_{-2}$ °C immediately followed by qualification or performance test, as applicable.
Qualification test - 2,000 hours at 70 $^{+3}_{-0}$ °C with 60 cycle rms, equal to peak working voltage, applied.
Dielectric withstanding voltage - Initial requirement.
Insulation resistance - Initial requirement.
Dissipation factor - Not greater than 150 percent of initial value.
Capacitance change - Shall not exceed ±8 percent of nominal value or 1 pf, whichever is greater, for characteristic B; ±3 percent of nominal value or 1 pf, whichever is greater, for characteristics D and F.
Performance check - 250 hours at conditions specified for qualification test.
Dielectric withstanding voltage - Initial requirement.
Insulation resistance - Initial requirement.
Dissipation factor - Not greater than 150 percent of initial value.
Capacitance change - Shall not exceed ±5 percent of nominal value or 1 pf, whichever is greater, for characteristic B; ±2 percent of nominal value or 1 pf, whichever is greater, for characteristics D and F.

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TABLE I. Style CM95.

Type designation 1/	Capacitance value (pf)	Peak working voltage (volts)	Rated current at frequencies (rms amps)				Characteristic available for capacitance tolerance ±5% (J)	Operating temperature range
			3.0 mc/sec	1.0 mc/sec	0.3 mc/sec	0.1 mc/sec		
CM95-X101JM1	100	35,000	12	6.2	2.2	0.82	B, D, F	M
CM95-X111JM1	110	35,000	13	6.8	2.4	0.91	B, D, F	M
CM95-X121JM1	120	35,000	13	7.5	2.7	1.0	B, D, F	M
CM95-X131JM1	130	35,000	13	8.2	3.0	1.1	B, D, F	M
CM95-X151JM1	150	35,000	15	9.1	3.6	1.2	B, D, F	M
CM95-X161JM1	160	35,000	15	9.1	3.9	1.3	B, D, F	M
CM95-X181JM1	180	35,000	16	10	4.3	1.3	B, D, F	M
CM95-X201JM1	200	35,000	16	11	4.7	1.5	B, D, F	M
CM95-X221JM1	220	35,000	16	11	5.1	1.6	B, D, F	M
CM95-X241JM1	240	35,000	18	12	5.6	1.8	B, D, F	M
CM95-X271JM1	270	35,000	18	13	6.2	2.0	B, D, F	M
CM95-X301JM1	300	35,000	18	13	6.8	2.0	B, D, F	M
CM95-X331JM1	330	35,000	20	13	6.8	2.2	B, D, F	M
CM95-X361JM1	360	35,000	20	15	7.5	2.4	B, D, F	M
CM95-X391JM1	390	35,000	20	16	8.2	2.7	B, D, F	M
CM95-X431JM1	430	35,000	22	16	8.2	3.0	B, D, F	M
CM95-X471JM1	470	35,000	22	18	9.1	3.0	B, D, F	M
CM95-X511JM1	510	35,000	22	18	9.1	3.0	B, D, F	M
CM95-X561JM1	560	35,000	24	20	10	3.3	B, D, F	M
CM95-X621JM1	620	35,000	24	20	11	3.6	B, D, F	M
CM95-X681JM1	680	35,000	24	22	11	3.9	B, D, F	M
CM95-X751JM1	750	35,000	27	22	12	4.3	B, D, F	M
CM95-X821JM1	820	35,000	27	24	12	4.3	B, D, F	M
CM95-X911JM1	910	35,000	27	24	13	4.7	B, D, F	M
CM95-X102JM1	1,000	35,000	30	27	15	5.1	B, D, F	M
CM95-X112JM1	1,100	35,000	30	27	15	5.6	B, D, F	M
CM95-X122JM1	1,200	35,000	30	30	16	6.2	B, D, F	M
CM95-X132JM1	1,300	35,000	33	30	16	6.2	B, D, F	M
CM95-X152JM1	1,500	35,000	33	33	18	6.8	B, D, F	M
CM95-X162JM1	1,600	35,000	33	33	18	7.5	B, D, F	M
CM95-X182JM1	1,800	35,000	33	36	20	7.5	B, D, F	M
CM95-W202JM1	2,000	30,000	36	39	20	8.2	B, D, F	M
CM95-W222JM1	2,200	30,000	36	39	22	9.1	B, D, F	M
CM95-W242JM1	2,400	30,000	39	43	22	10	B, D, F	M
CM95-W272JM1	2,700	30,000	39	43	24	10	B, D, F	M
CM95-W302JM1	3,000	30,000	39	47	24	11	B, D, F	M
CM95-W332JM1	3,300	30,000	39	47	27	11	B, D, F	M
CM95-V362JM1	3,600	25,000	43	51	27	12	B, D, F	M
CM95-V392JM1	3,900	25,000	43	51	30	15	B, D, F	M
CM95-V432JM1	4,300	25,000	43	56	30	15	B, D, F	M
CM95-V472JM1	4,700	25,000	43	56	33	15	B, D, F	M
CM95-V512JM1	5,100	25,000	43	56	33	15	B, D, F	M
CM95-U562JM1	5,600	20,000	47	62	36	16	B, D, F	M
CM95-U622JM1	6,200	20,000	47	62	36	16	B, D, F	M
CM95-T682JM1	6,800	15,000	47	62	39	18	B, D, F	M
CM95-T752JM1	7,500	15,000	47	68	39	18	B, D, F	M
CM95-T822JM1	8,200	15,000	47	68	43	18	B, D, F	M
CM95-T912JM1	9,100	15,000	51	68	43	20	B, D, F	M
CM95-T103JM1	10,000	15,000	51	75	47	20	B, D, F	M

1/Complete type designation will include an additional symbol to indicate characteristic.

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Continuation test - Units that have been subjected to the 250 hour test shall be continued for an additional 1,750 hours and meet the requirements for the qualification test.

Qualification inspection:
Qualification inspection (Group E) is required.

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Army - EL
Navy - AS, OS, EC, SH
Air Force - 11, 17, 99

User activities:
Army -
Navy - MC, CG
Air Force - 14, 19

Preparing activity:
Navy - EC

Agent:
DSA - ES

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	☐ OTHER *(Specify)*: _____

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a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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